

Text Level 3: Chemistry

visual context

(Completely Irrelevant)



Question: How many chemically distinct hydrogen atoms are there on 4?

Options:

- A: 4
- B: 7
- C: 8
- D: 10

textual context

7-(tert-butoxy)bicyclo[2.2.1]hepta-2,5-diene is combined with 2 equivalents of 5,6-bis(dibromomethyl)cyclohexa-1,3-diene and sodium iodide at elevated temperature, forming product 1. 1 is treated with aqueous sulfuric acid, forming 2. 2 is treated with SO₃ and pyridine in DMSO, forming 3. 3 is heated at 150°C, forming final product 4.

GPT 5.1: Let's analyze the sequence step by step and infer the structure of product 4, then count its distinct hydrogens.

1. Starting material and the cycloadduct (product 1): 7-(tert-butoxy)bicyclo[2.2.1]hepta-2,5-diene is essentially a norbornadiene system with a tert-butoxy substituent at the bridgehead (C7). 5,6-Bis(dibromomethyl)cyclohexa-1,3-diene reacts twice (two equivalents) with the diene in a Diels–Alder-type fashion (facilitated by NaI, which transforms dibromomethyl into exocyclic methylene or related reactive intermediates). This gives a rigid polycyclic adduct (1). For the purposes of the later steps, key features are that the tert-butoxy substituent and the original norbornane core survive.

2. Acidic treatment (1 → 2): Aqueous H₂SO₄ hydrolyzes the tert-butoxy group to give the corresponding tertiary alcohol, or more plausibly the tert-butyl cation path leads to a hydroxyl substituent on the bridgehead carbon (loss of t-BuOH).

3. SO₃ / pyridine in DMSO (2 → 3): This is a variant of the Parikh–Doering / Swern-type oxidation environment; under these conditions a primary or secondary alcohol is oxidized to carbonyl. Thus the bridgehead hydroxyl in 2 is oxidized to a carbonyl (most consistently a ketone).

4. Heating 3 at 150 °C (3 → 4): Heating a bridgehead carbonyl in this highly strained polycyclic system promotes retro-Diels–Alder or related fragmentation, simplifying the skeleton. The net outcome after these rearrangements is a more symmetric polycyclic ketone with a limited number of unique positions.

Because the question is about number of chemically distinct hydrogens (i.e., different sets under the molecular symmetry), what matters is the final symmetry of this rigid framework. The reorganized polycyclic ketone 4 is quite symmetrical; analysis of its nonequivalent carbon positions (considering mirror planes and C₂ axes) shows that its protons fall into 7 different environments. Thus, the number of chemically distinct hydrogen atoms on 4 is:

7

Ground Truth: A:4